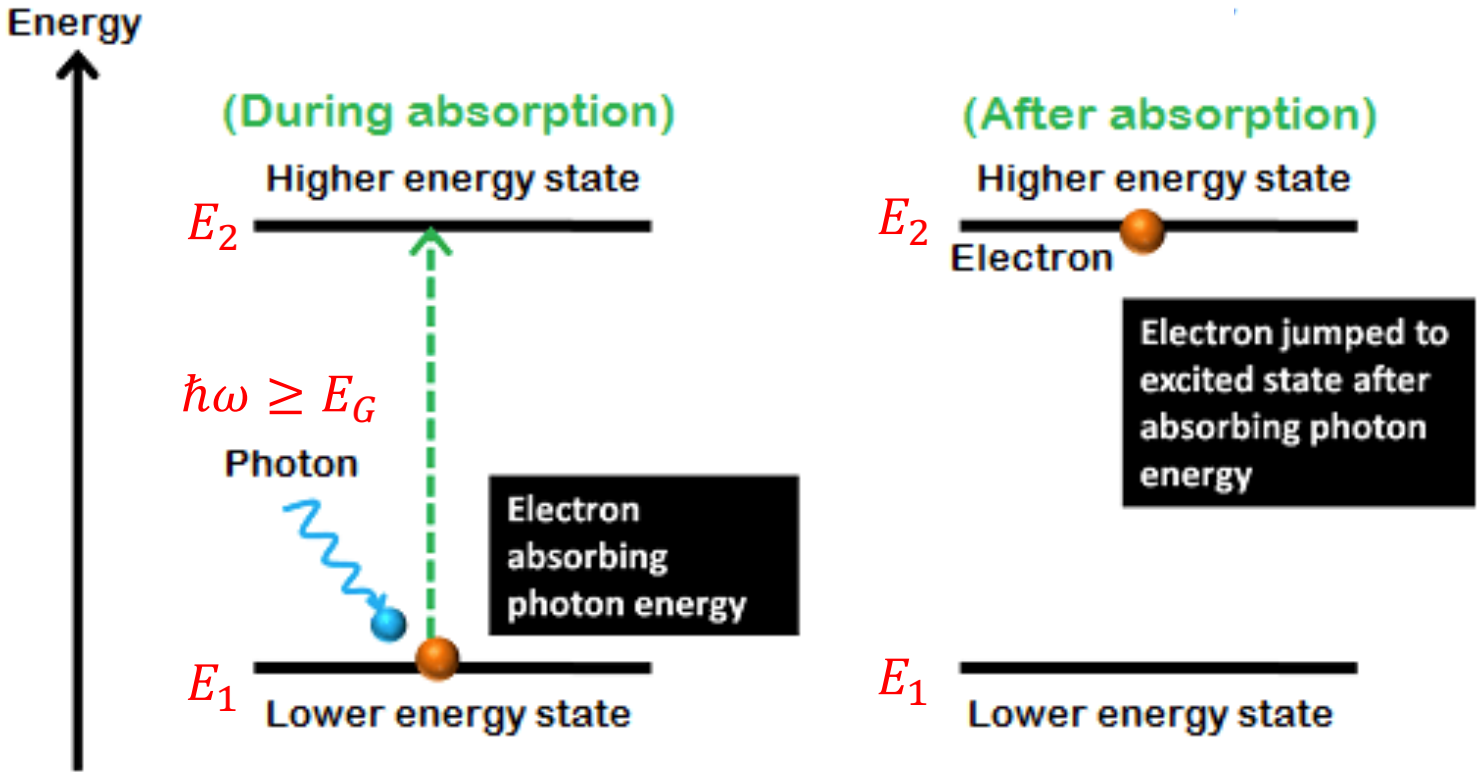
Photodetectors

The following subjects will be covered

- Modes of Photodiode Operation.
- Photodiode Circuit Analysis.
- Photo-transistors.

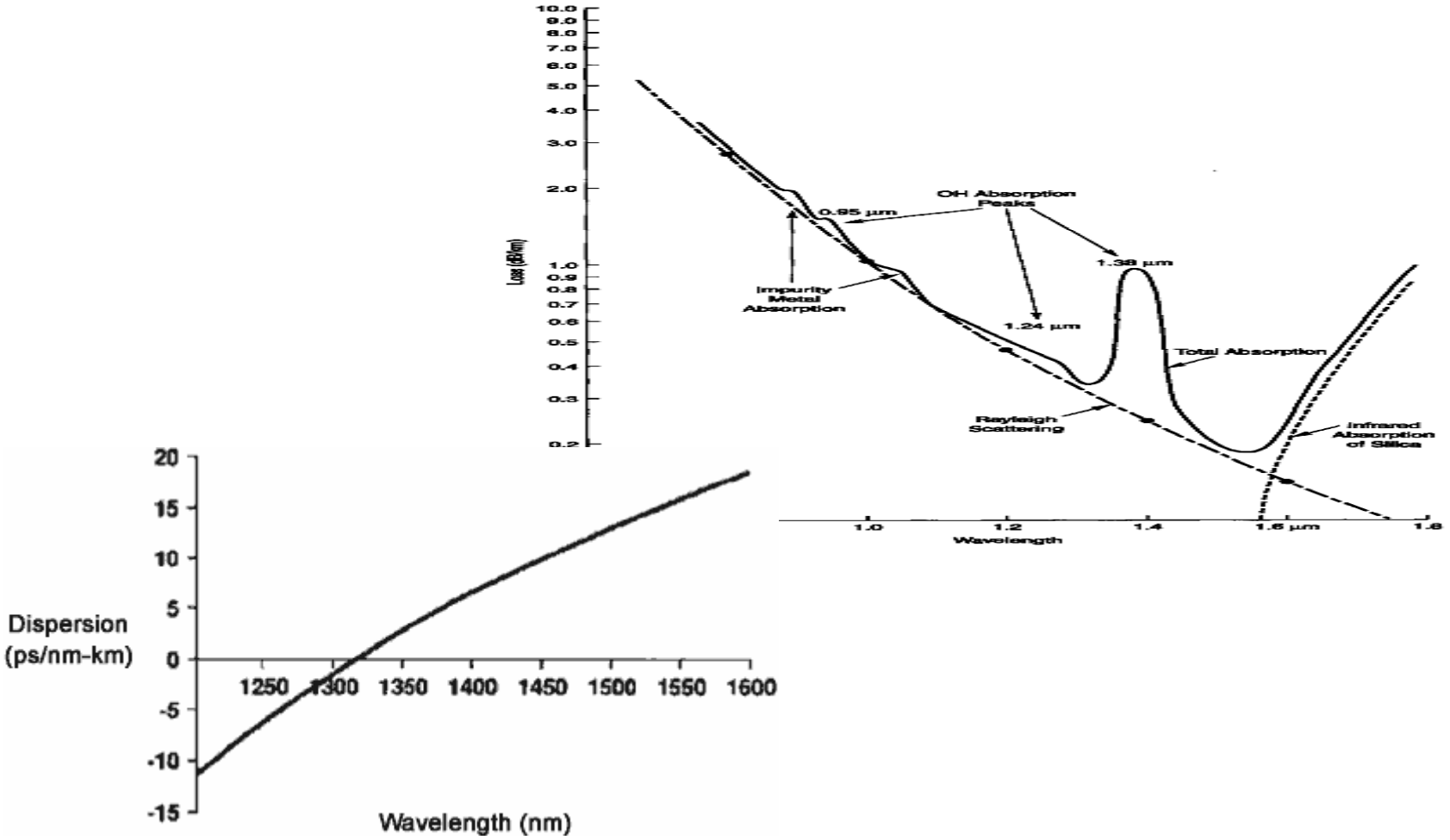
Experiment 1

Absorption



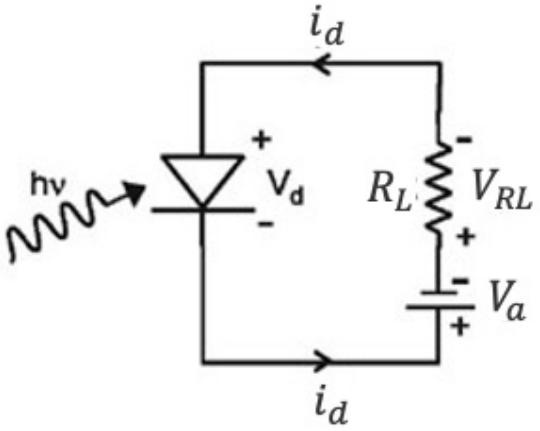
Electron-hole Generation

Optical Fiber Loss Curve

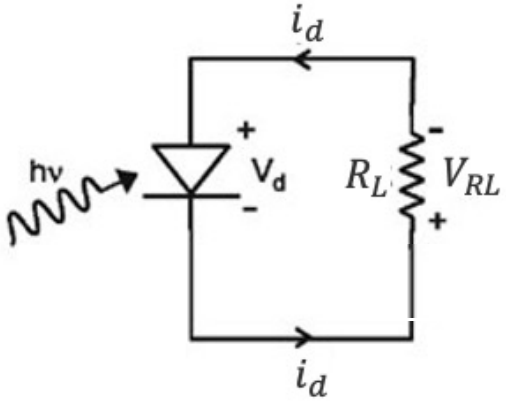


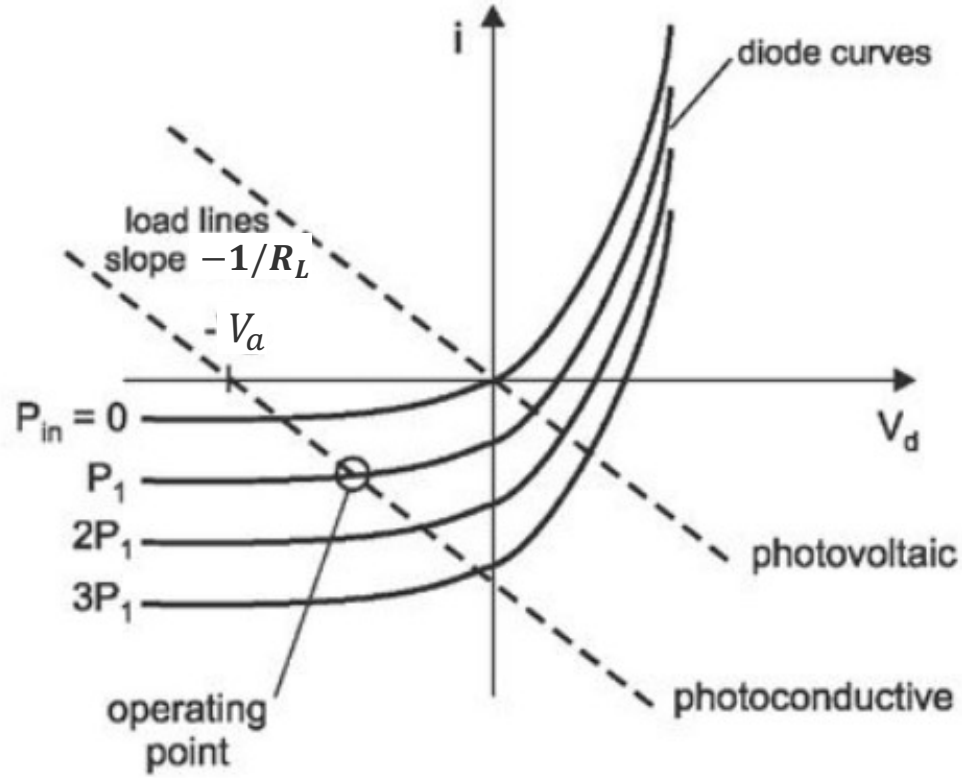
Photodiode Modes of Operation

Photoconductive Mode



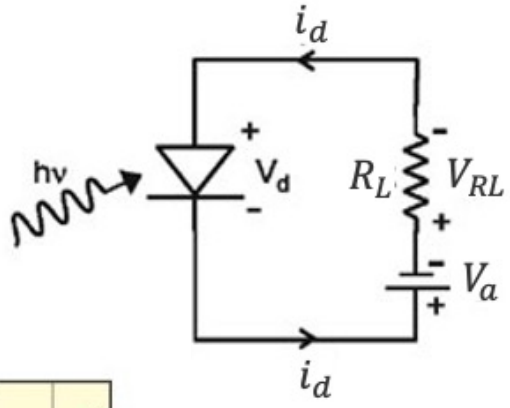
Photovoltaic Mode

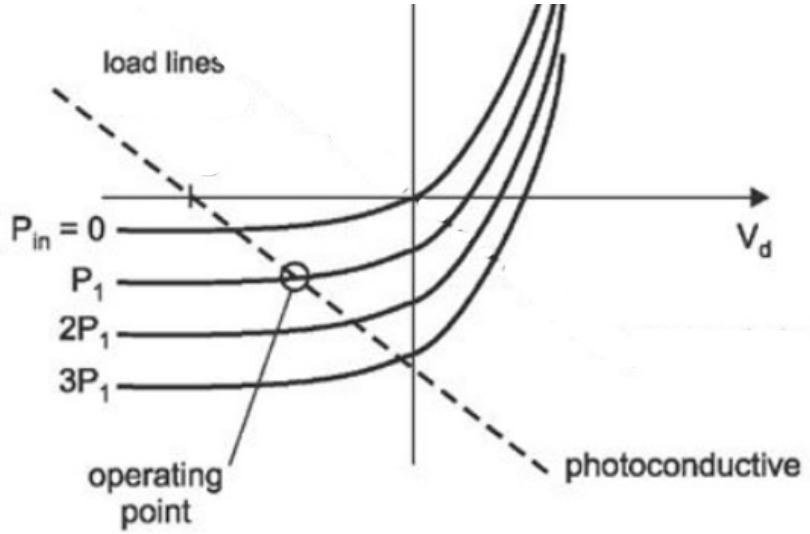


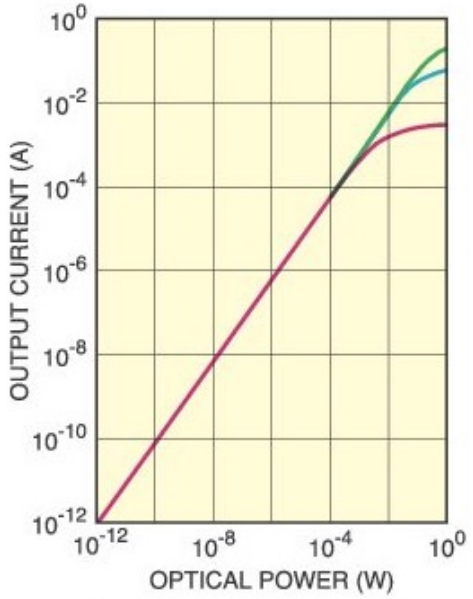


Photodiode Circuit Analysis

The current in the circuit is composed of the diode (dark) current and the photocurrent.







Photodiode dynamic range is determined by the linear response range. It's the useful range of sensor response. (Given in dB)

Dark current

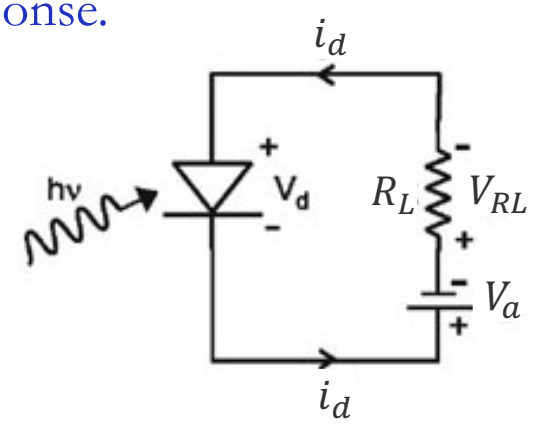
$$i_d = i_s\,(e^{V_d/nV_T} - 1) \approx -i_s$$

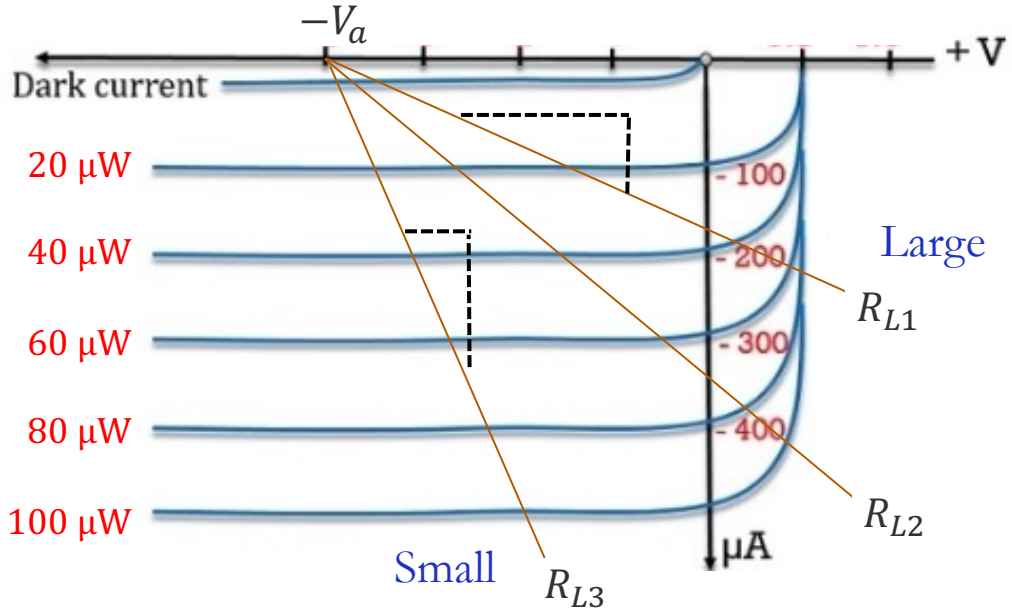
Total current

$$i_d = i_s\,(e^{V_d/nV_T} - 1) - i_p$$

Photodiode Circuit Analysis

The choice of load resistance affects the sensitivity and the dynamic range of the photodiode response.





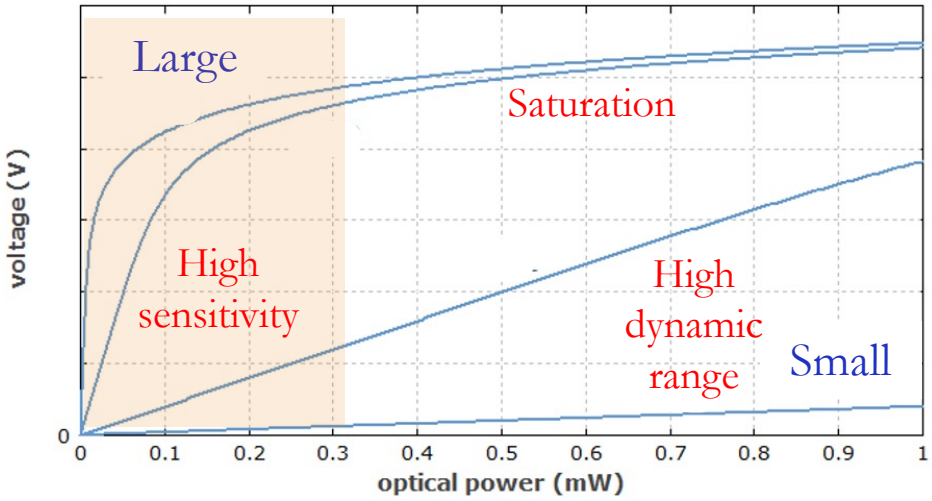
Find the load line (circuit) equation.
KVL for the circuit.

$$V_a + V_{RL} + V_d = 0$$

$$V_a + i_d R_L + V_d = 0$$

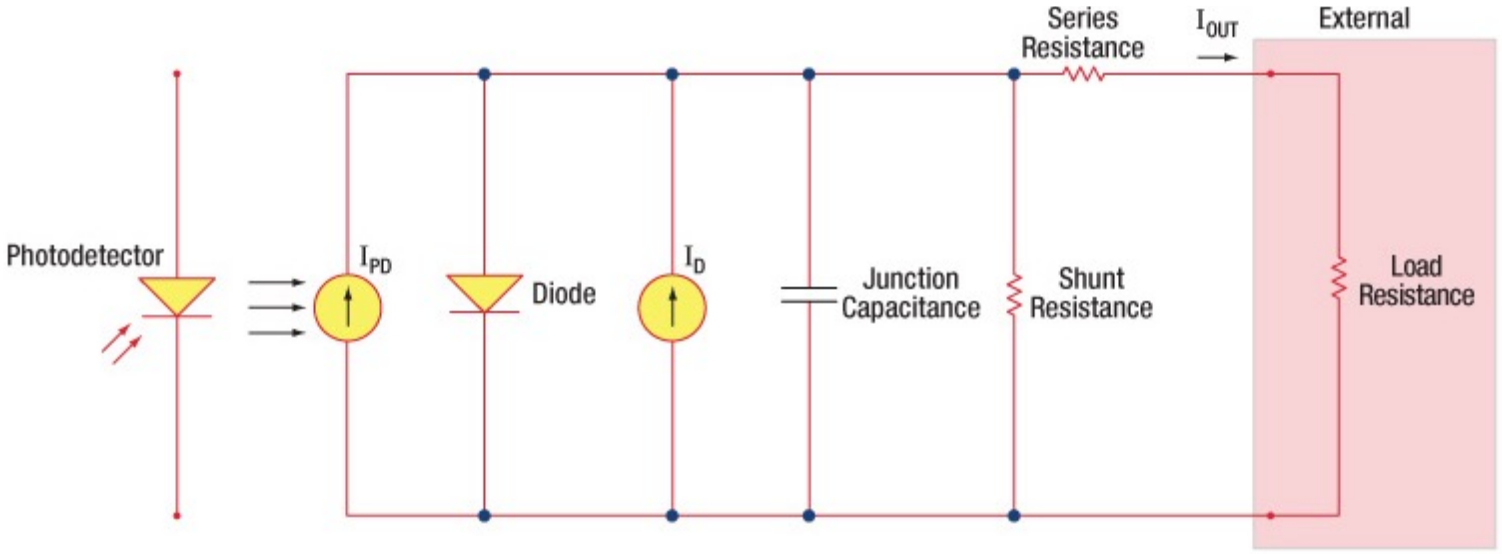
$$i_d = -\frac{V_a}{R_L} - \frac{V_d}{R_L}$$

Choice of R_L depends on application!



Photodiode Equivalent Circuit

The equivalent circuit of the photodiode is shown below.



The photodiode is represented by the following circuit elements.

A current source representing the generated photo current, I_{PD}

A current source representing the dark current , $I_D = I_S$.

An ideal diode.

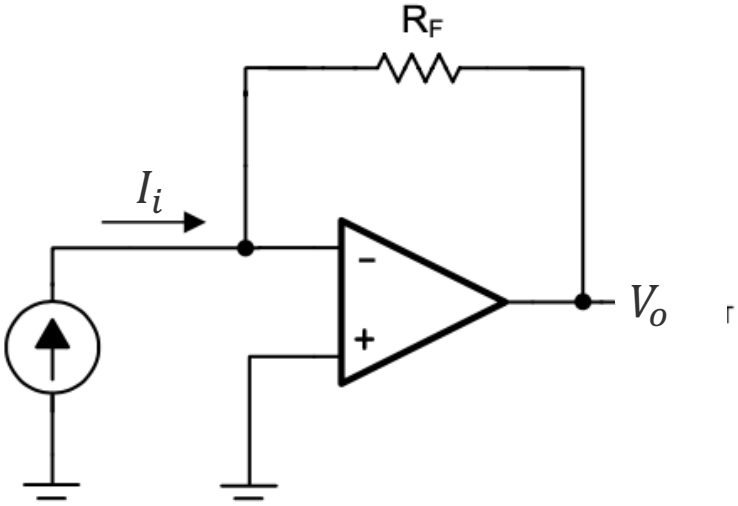
A parallel (shunt) capacitor representing the junction capacitance, C_j

A parallel (shunt) resistance representing the depletion region resistance, R_{SH}

A series resistance representing the contacts resistance, R_S

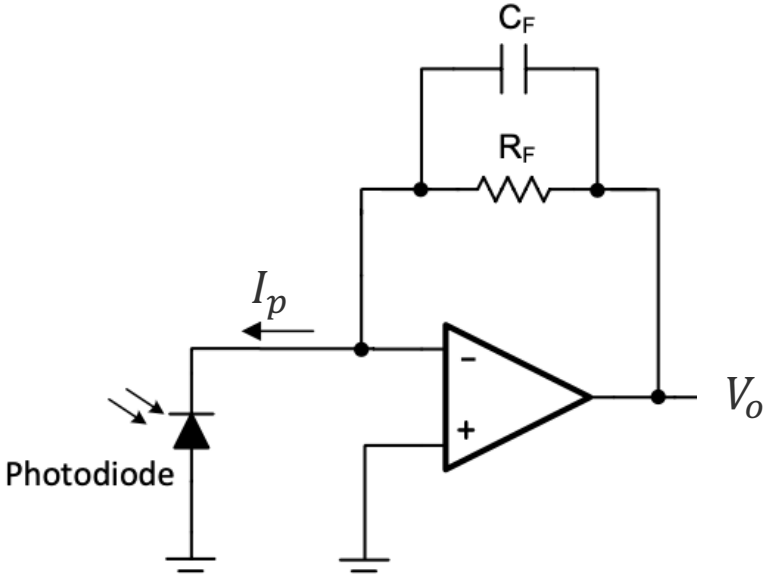
Photodiode Circuit Analysis

A trans-impedance amplifier is commonly used to amplify and set the bandwidth of operation.



A basic trans-impedance op-amp circuit

$$V_o = -I_i R_F$$

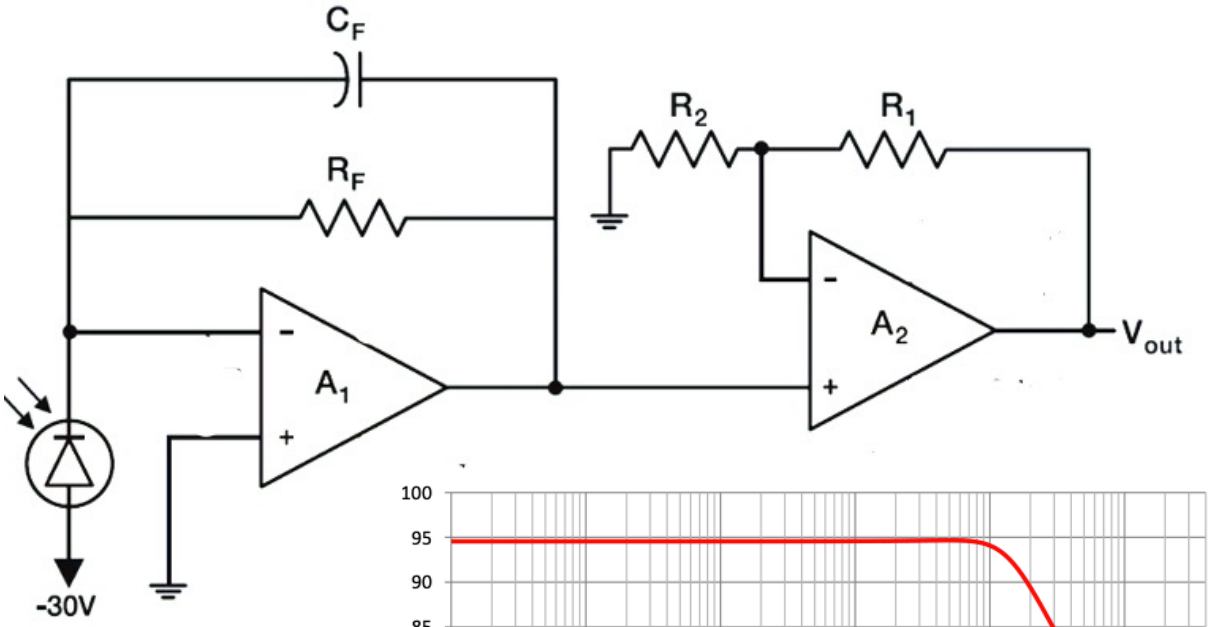


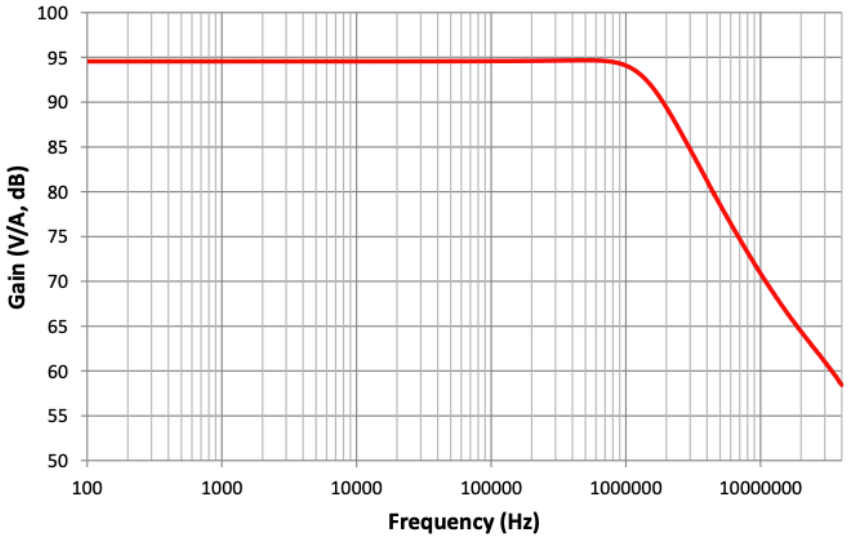
A trans-impedance op-amp circuit with feedback capacitor

$$V_o = I_p R_F$$

Photodiode Circuit Analysis

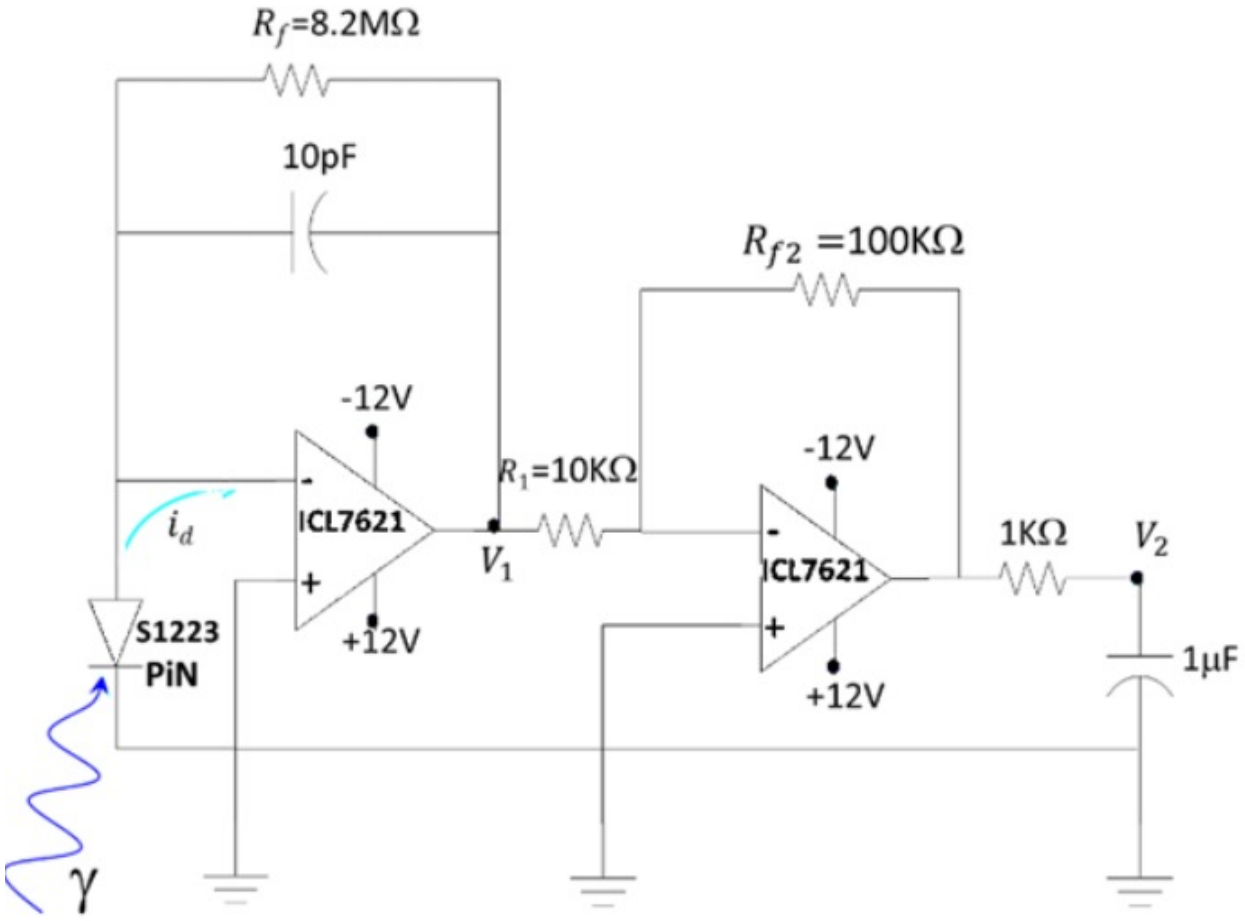
A trans-impedance amplifier is commonly used to amplify and set the bandwidth of operation.





Photodiode Circuit Analysis

A trans-impedance amplifier is commonly used to amplify and set the bandwidth of operation.



Photodiode Applications

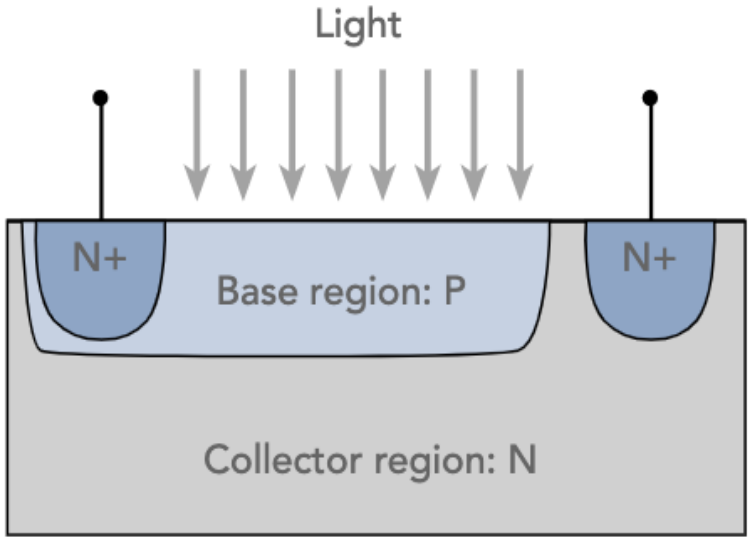
Basic Applications

- Optical communications. Photodiodes can provide high sensitivity, high speed and large bandwidth and dynamic range.
- Thermal imaging and temperature measurements. A combination of designs can be used for this purpose (photodetectors and thermopiles, for example).
- Consumer electronics devices like smoke detectors, compact disc players, and televisions and remote controls in VCRs.
- Measurement of the intensity of light in science and industry experimental research and production. Generally, they have an enhanced, more linear response than general photoconductors and phototransistors.
- Medical applications like instruments to analyze samples, detectors for computed tomography, blood gas monitors, thermometers.
- Lighting regulation. Photodiodes can provide a sensitive feedback to regulate the light emitter output.

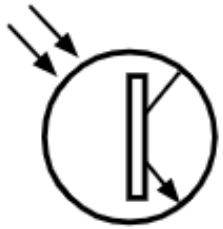
The Photo-Transistor

Main Features

- Transparent outside cover for illumination.
- Larger base and collector area for light absorption.
- When appropriately biased (active mode), can provide higher gain and sensitivity than a photodiode.
- Can easily be incorporated into an integrated circuit.
- Can offer a reasonable speed, but not used for high-speed applications.
- Provide electrical isolation.
- Mainly used as switches and sensors.
- No need for a gain circuit when configured as an amplifier.



- Lower noise level than APDs.
- Small bandwidth (< 250 kHz) because of junction capacitance.
- High sensitivity to temperature variations.



NPN Phototransistor

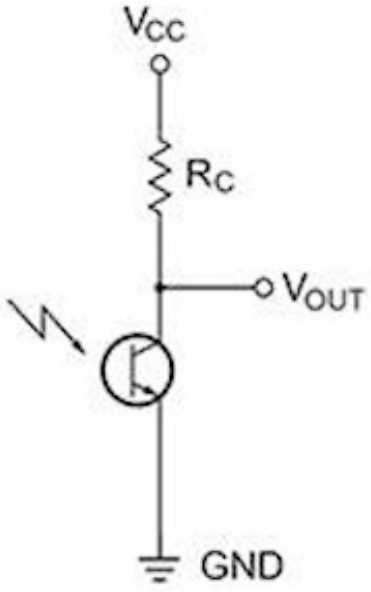
Photo-Transistor Applications

Basic Applications

- Opto-isolators: the phototransistor is used as a light sensor, the light emitter being relatively close, but at a different potential. The physical gap between the light emitter and detector provides a considerable degree of electrical isolation.
- Position sensing: in this application the opto-isolator can be used to detect the position of a moving element, often the moving element has a light or interrupts a beam of light which the phototransistor detects.
- Security systems: phototransistor can be used in many ways in security systems, often detecting whether a beam of light is present or has been broken by an intruder.
- Coin/item counters: phototransistor can be used in coin and other counting applications. A beam of light is interrupted each time a coin or other item passes a given point. The number of times the beam is interrupted equals the number of coins or objects to be counted.

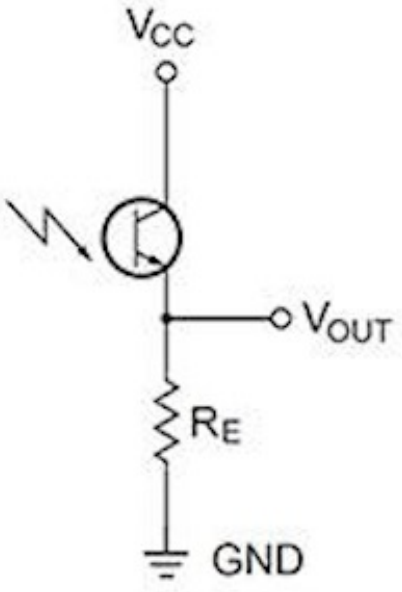
Photo-Transistor Circuits

Common-Emitter configuration



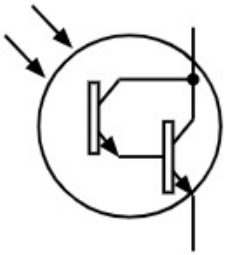
The linear region of this configuration is limited. So, it is mainly used for switching.

Common-Collector configuration



It can be used as an opposite switch to the CE configuration.

Darlington configuration



Enhanced sensitivity

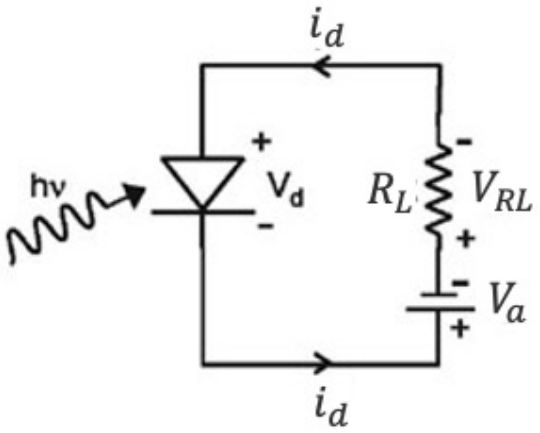
Photovoltaic Devices

The following subjects will be covered

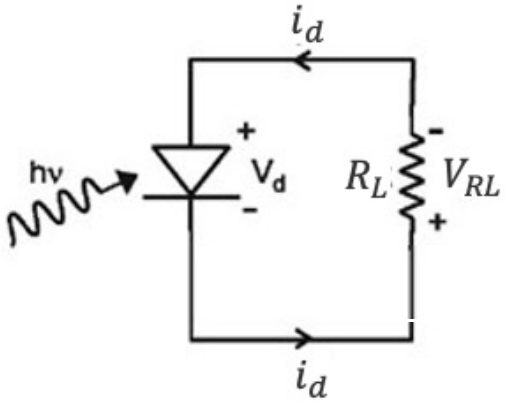
- Photovoltaic Mode of Operation.
- Basic Solar Cell Structure.
- Solar Radiation Fundamentals.
- Solar Cell Efficiency Analysis.
- Solar Cell Circuit Analysis.

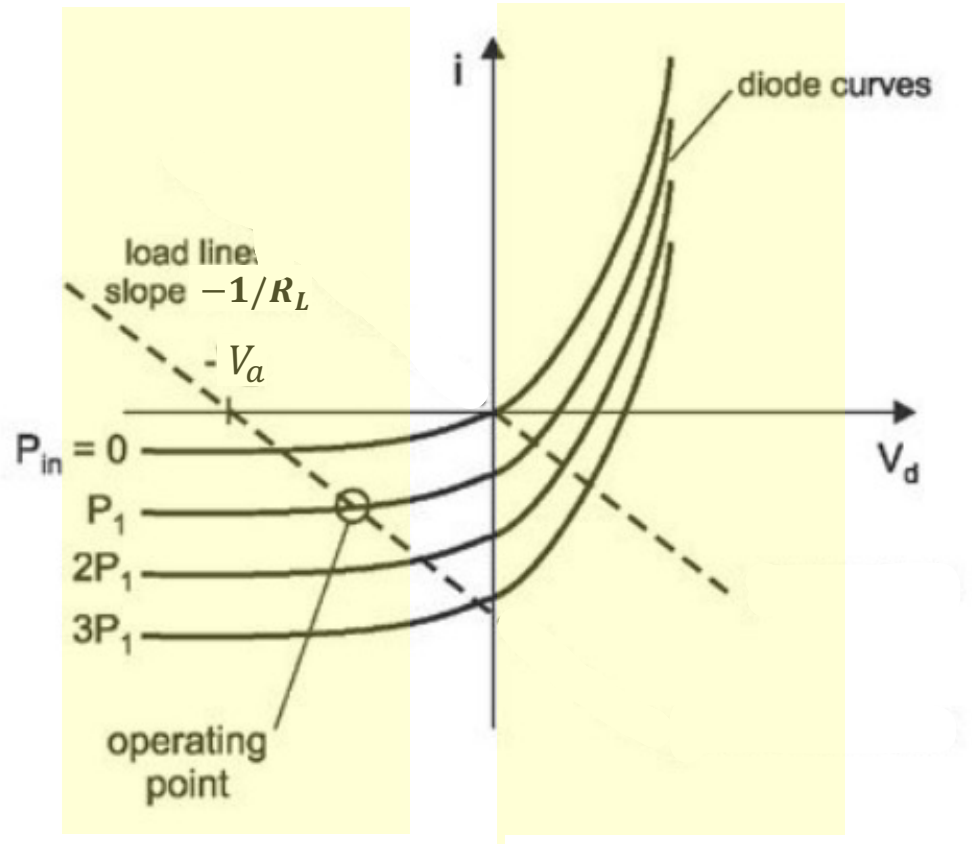
Photodiode Modes of Operation

Photoconductive Mode



Photovoltaic Mode

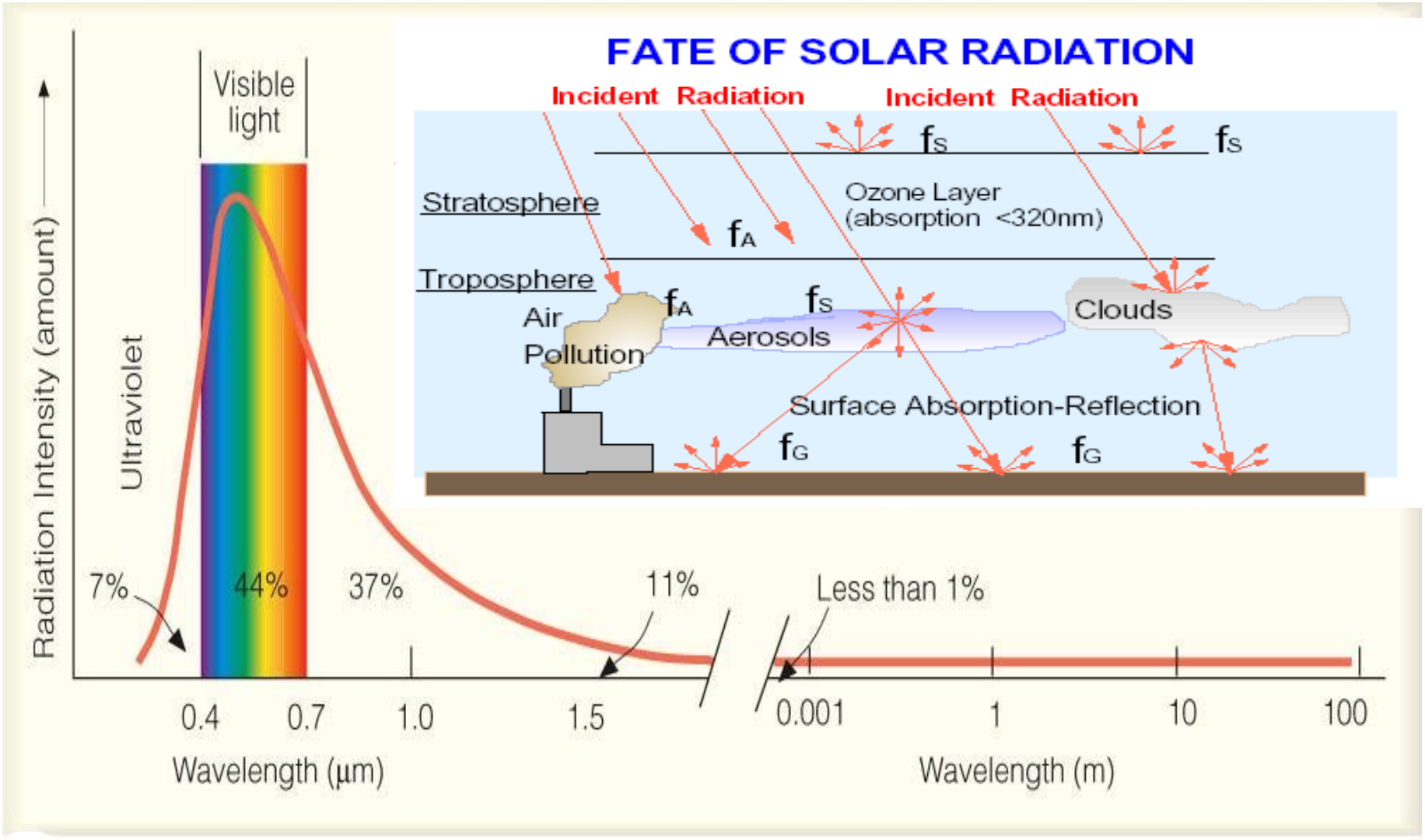




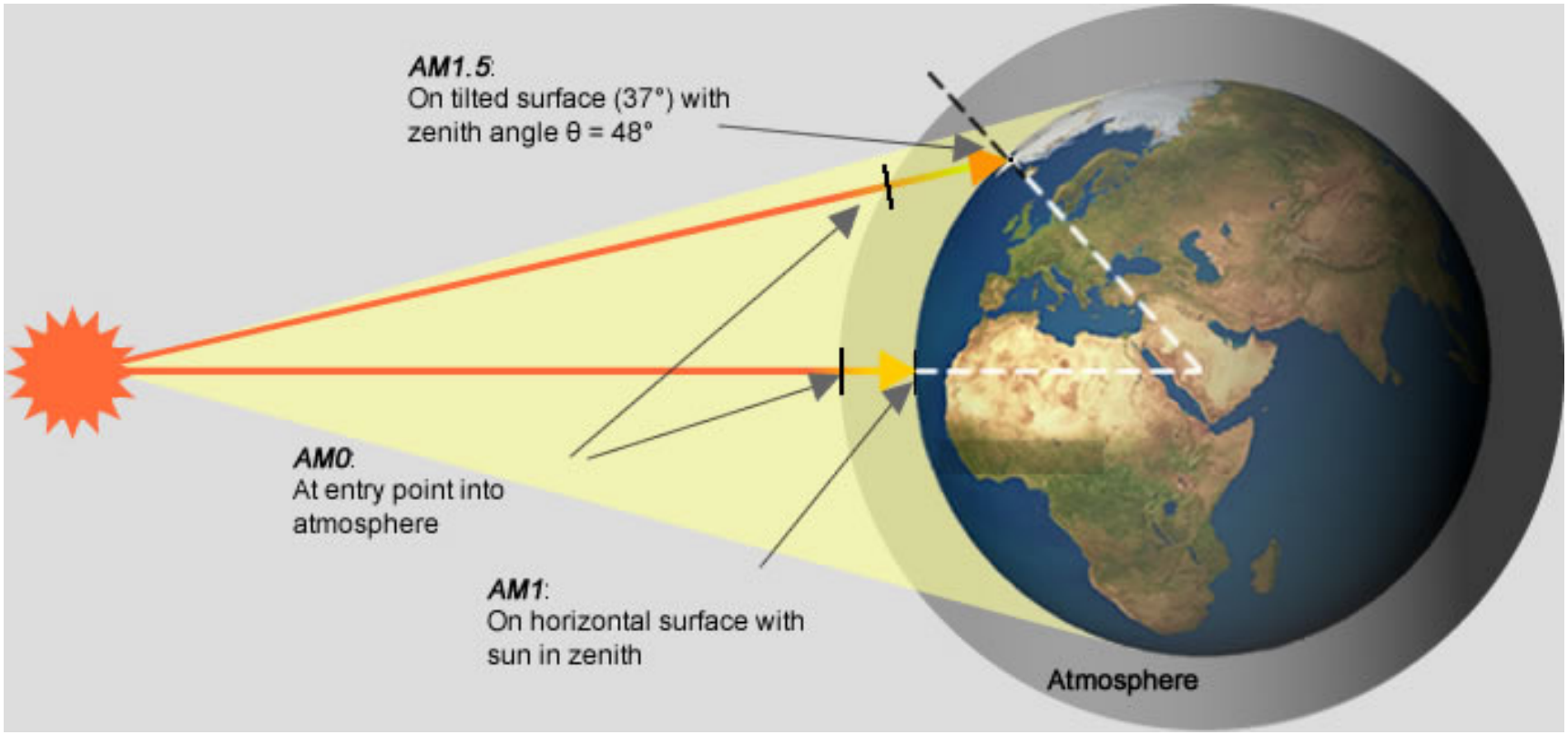
Photoconductive Mode

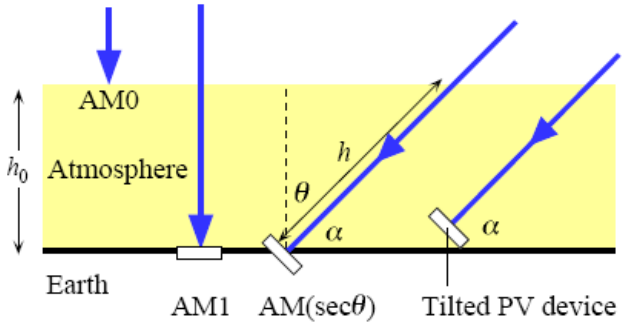
Photovoltaic Mode

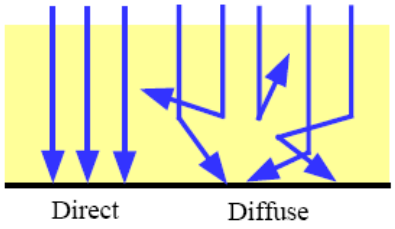
Solar Spectrum



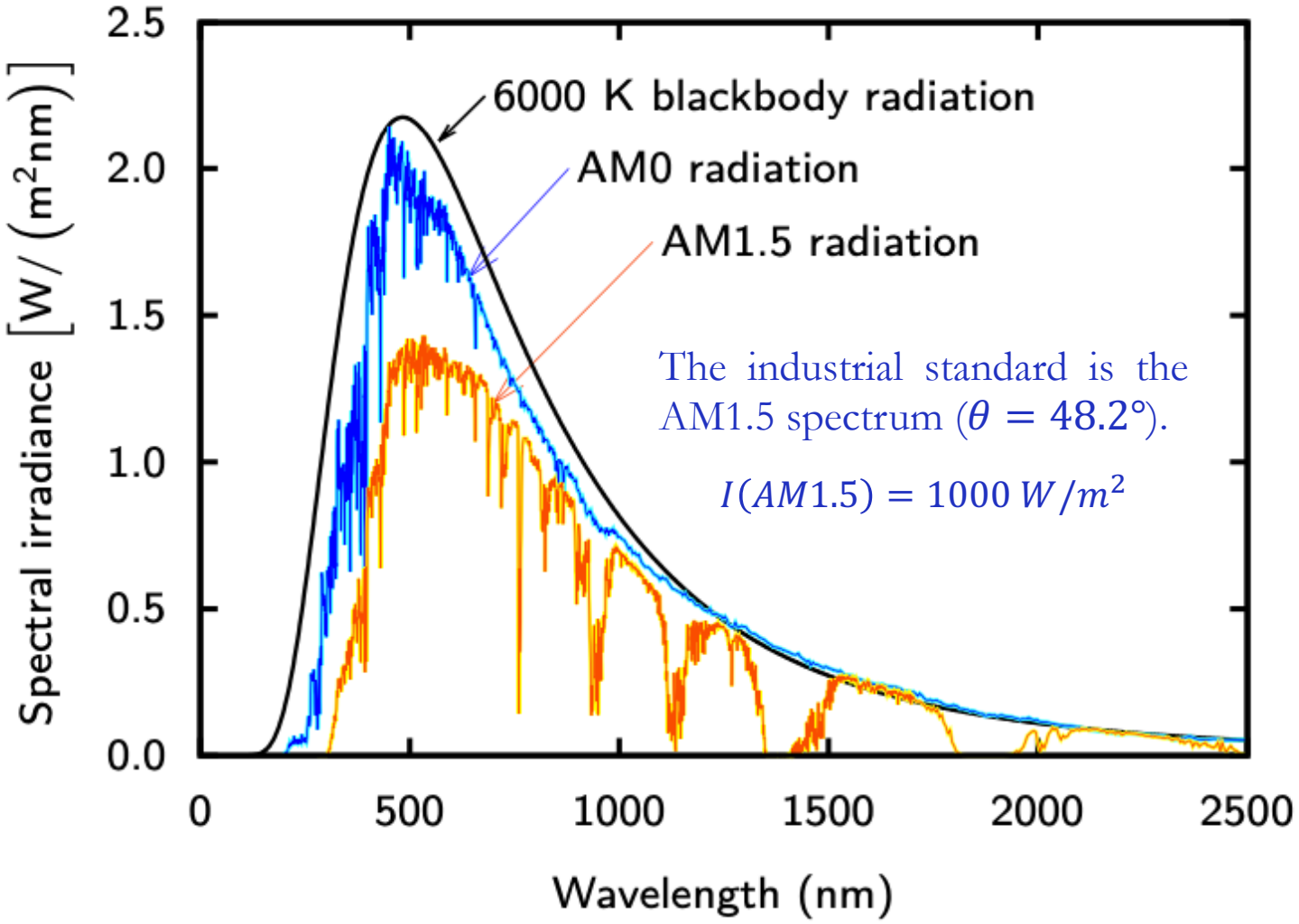
Solar Spectrum



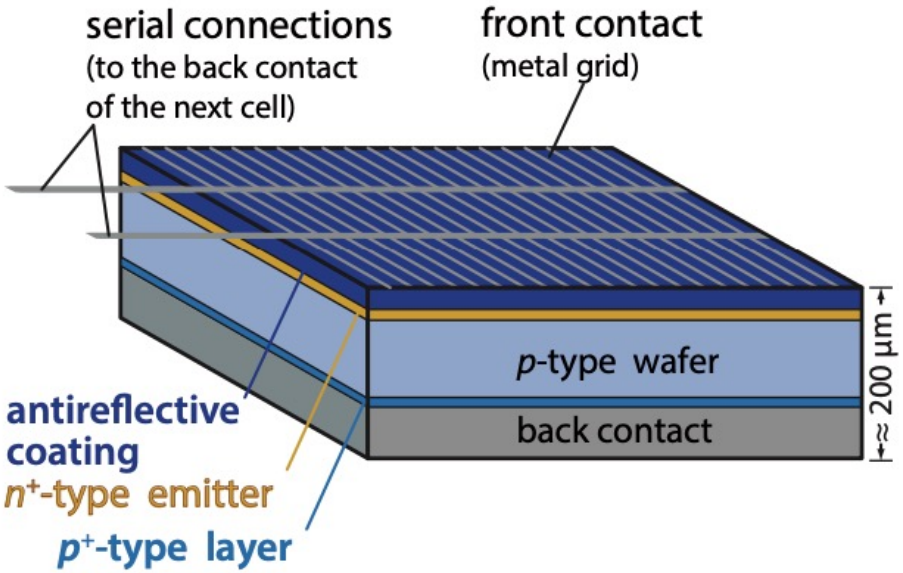


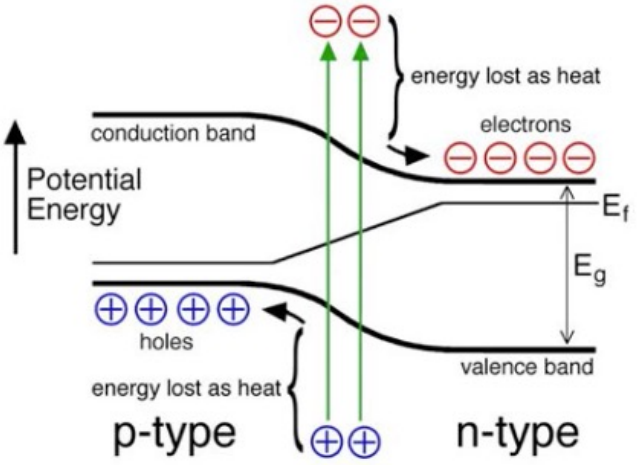


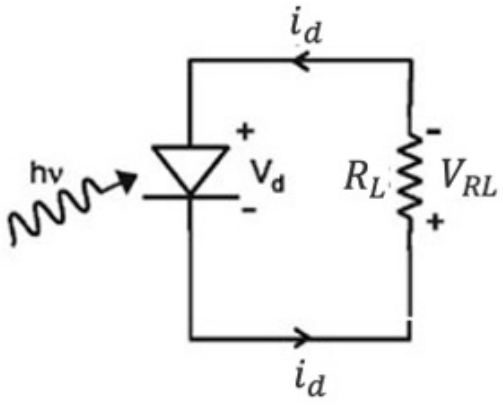
Solar Spectrum



Principle of Operation of PN Solar Cells







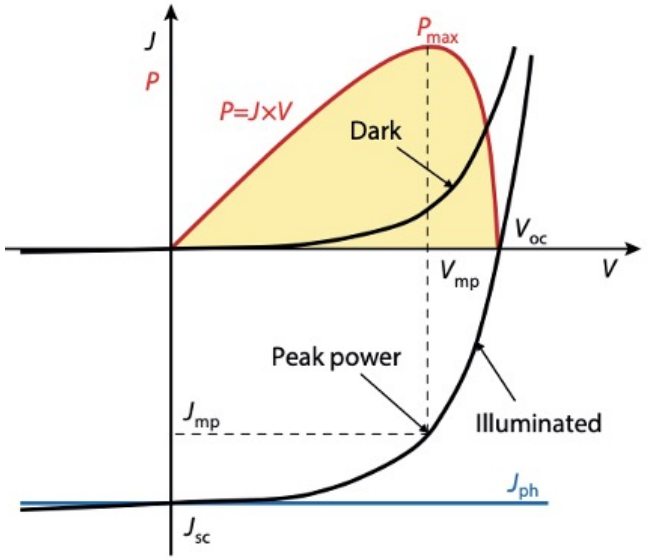
The conversion efficiency is characterized by:

1) Open-circuit voltage.

V_{oc}

2) Short-circuit current

I_{sc}



Design Aspects: Losses and Efficiency

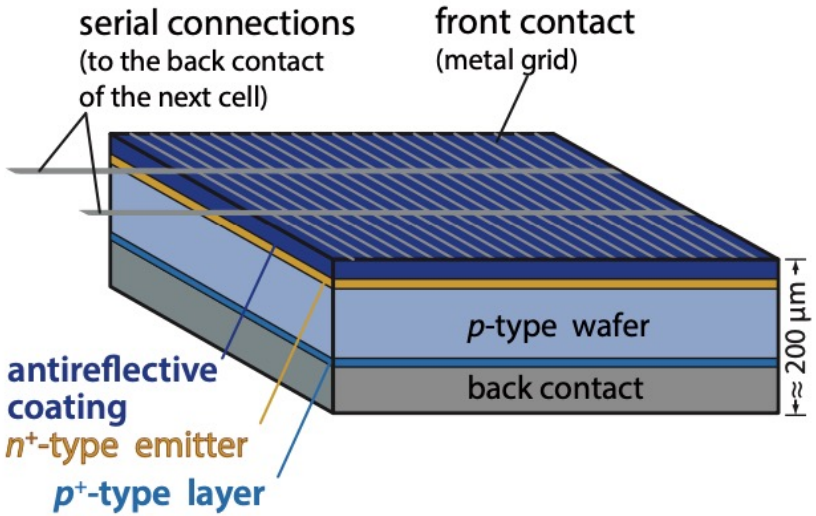
Factors affecting Open-circuit voltage V_{oc} and Short-circuit current I_{sc}.

Recombination.

Bulk Recombination because of defects and impurities. Also because of short minority lifetime.

Surface Recombination because of surface roughness and surface states (dangling bonds).

Fabrication process needs to be perfected.



Design Aspects: Losses and Efficiency

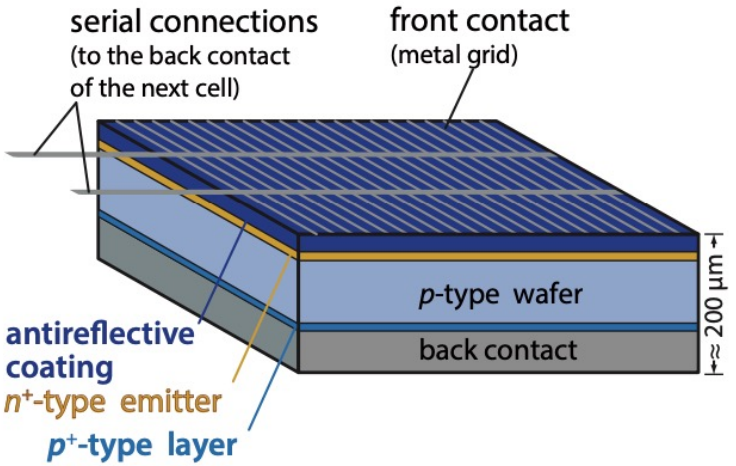
Factors affecting Open-circuit voltage V_{oc} and Short-circuit current I_{sc}.

Reflection and Transmission-through.

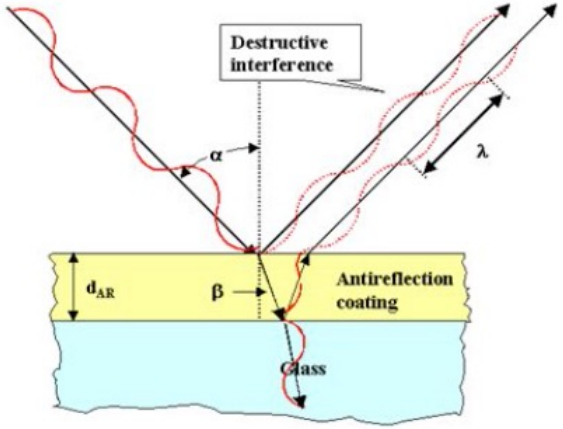
Free area ratio. $\frac{A_{free}}{A_{tot}}$

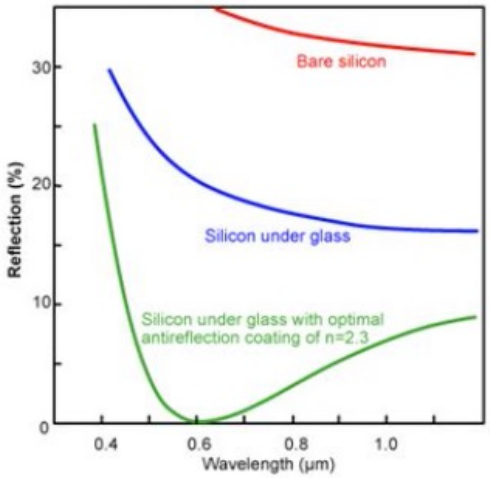
Fresnel Reflection. $(1 - R)$

Transmission-through. Device thickness.



- Electrode grid management.
- Anti-reflection coating.
- Back mirror.
- Surface texturing.





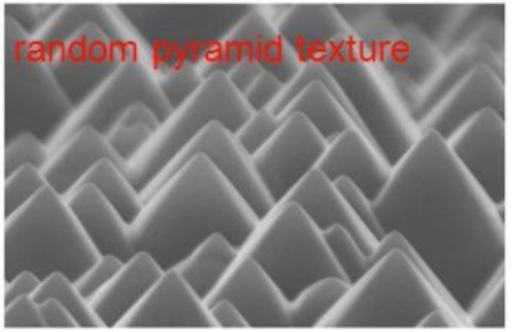
Design Aspects: Losses and Efficiency

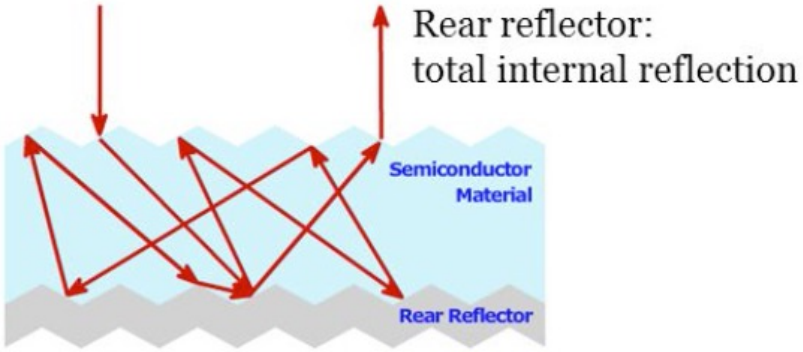
Factors affecting Open-circuit voltage V_{oc} and Short-circuit current I_{sc}.

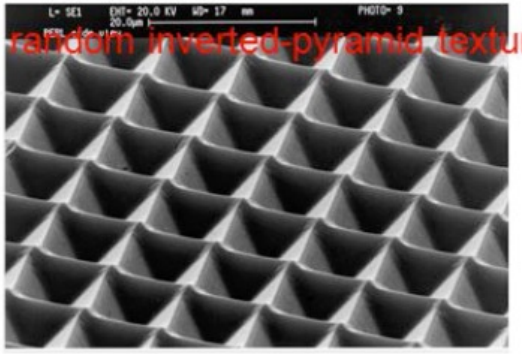
Reflection and Transmission-through.

Surface Texturing.









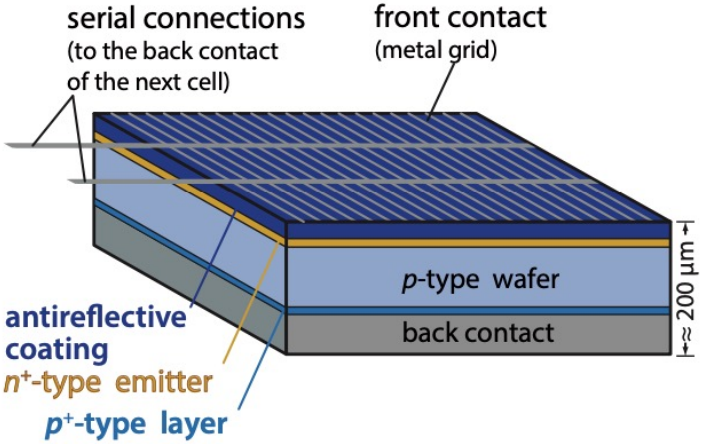
Design Aspects: Losses and Efficiency

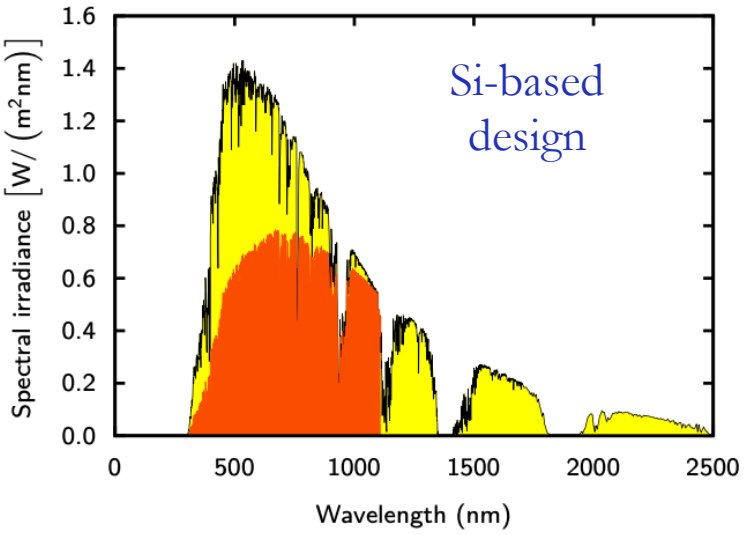
Factors affecting Open-circuit voltage V_{oc} and Short-circuit current I_{sc}.

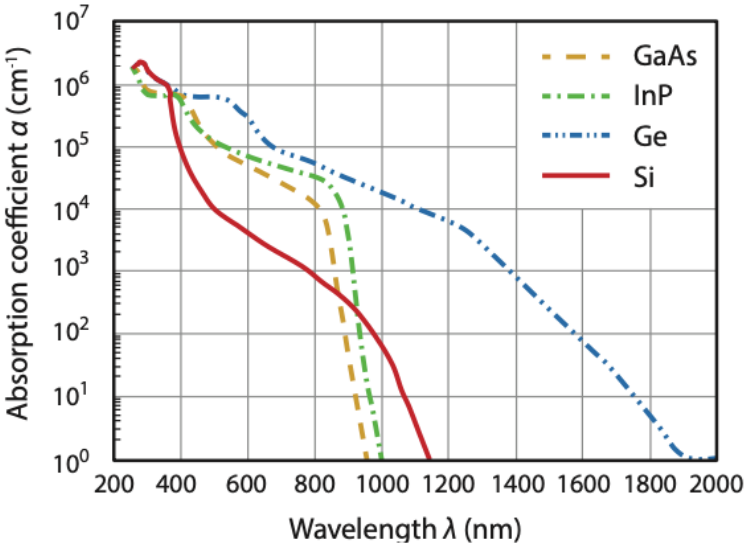
Partial Spectrum Absorption.

Limited material absorption (cutoff wavelength)

- Multi-material (bandgap) design.





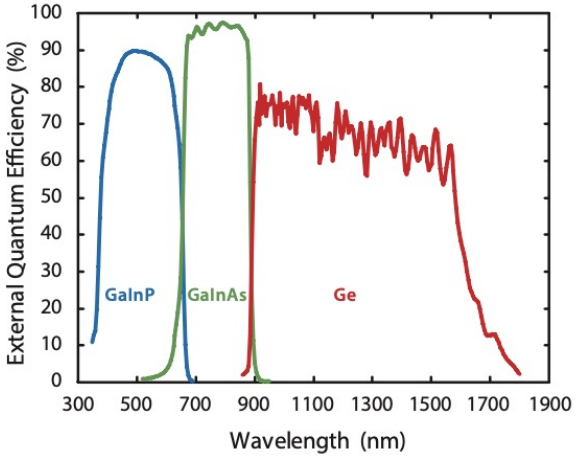


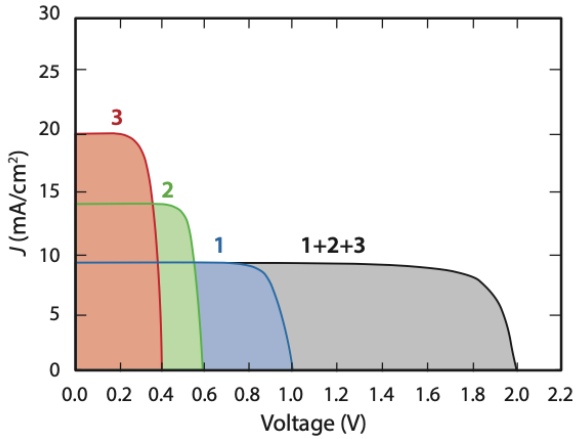
Design Aspects: Losses and Efficiency

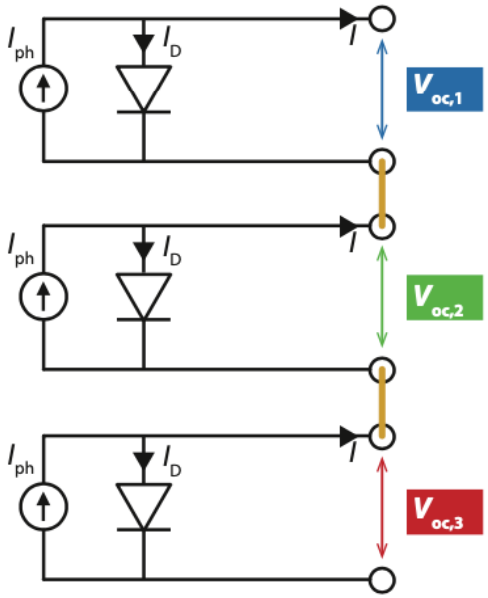
Factors affecting Open-circuit voltage V_{oc} and Short-circuit current I_{sc}.

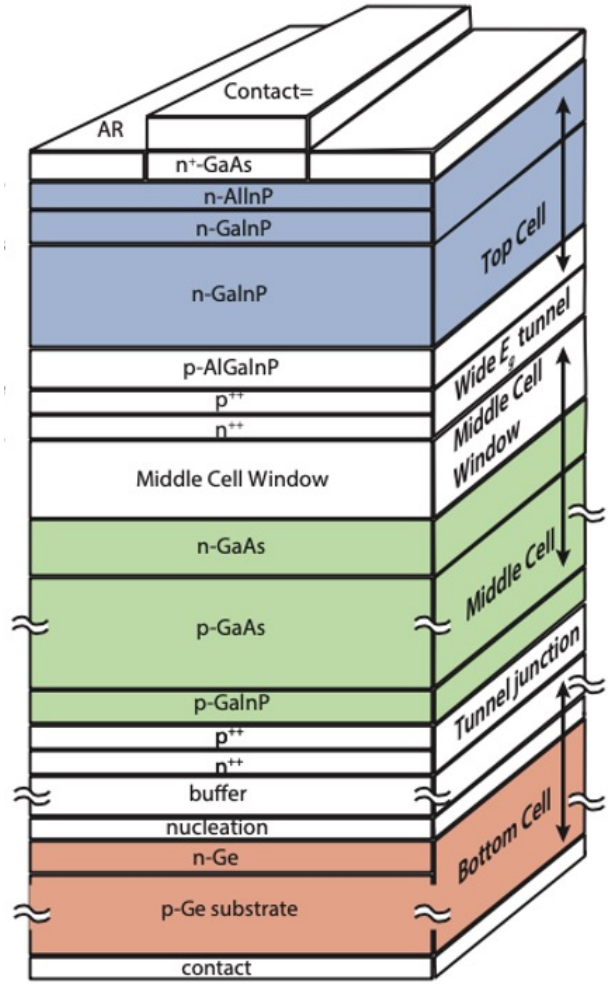
Partial Spectrum Absorption.

Multi-material (bandgap) design









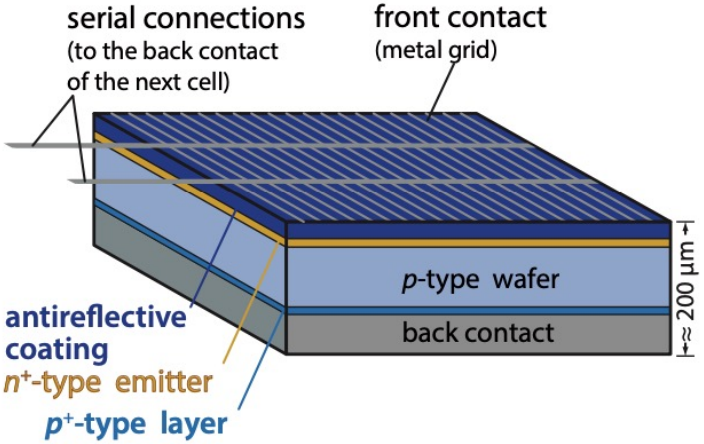
Design Aspects: Losses and Efficiency

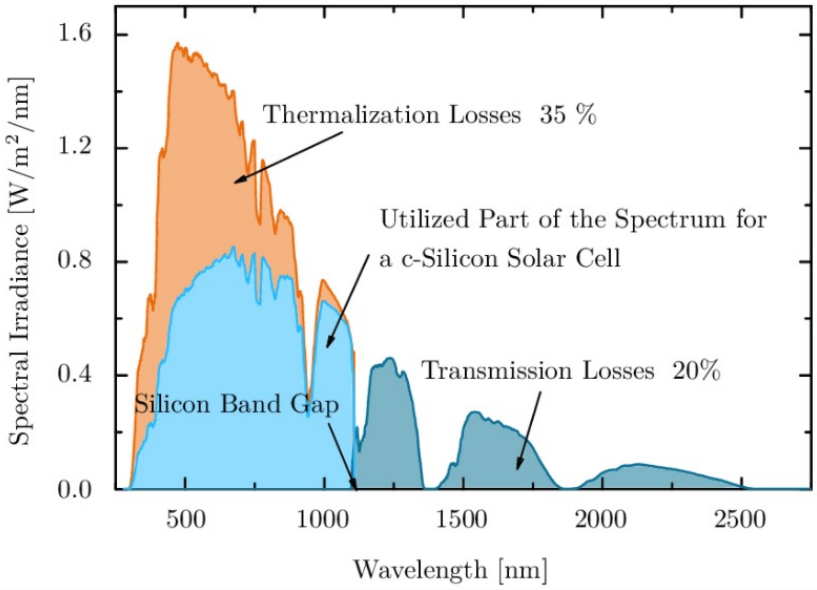
Factors affecting Open-circuit voltage V_{oc} and Short-circuit current I_{sc}.

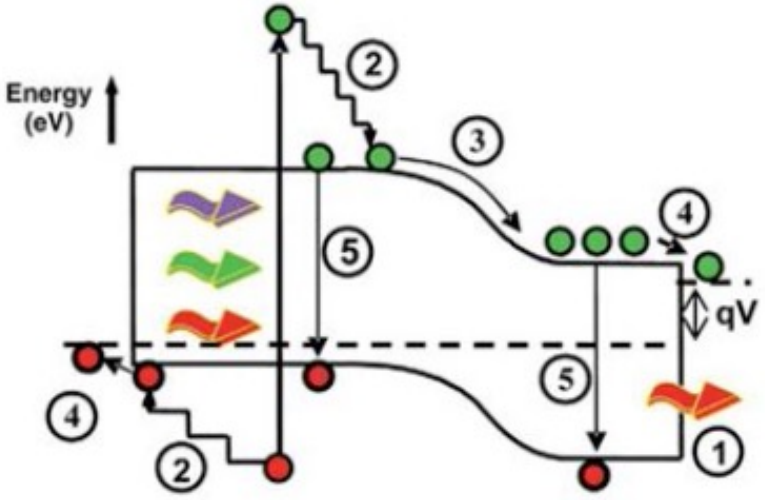
Thermalization.

Extra photon energy in converted to heat ($\hbar\omega > E_g$).

- Multi-material (bandgap) design.



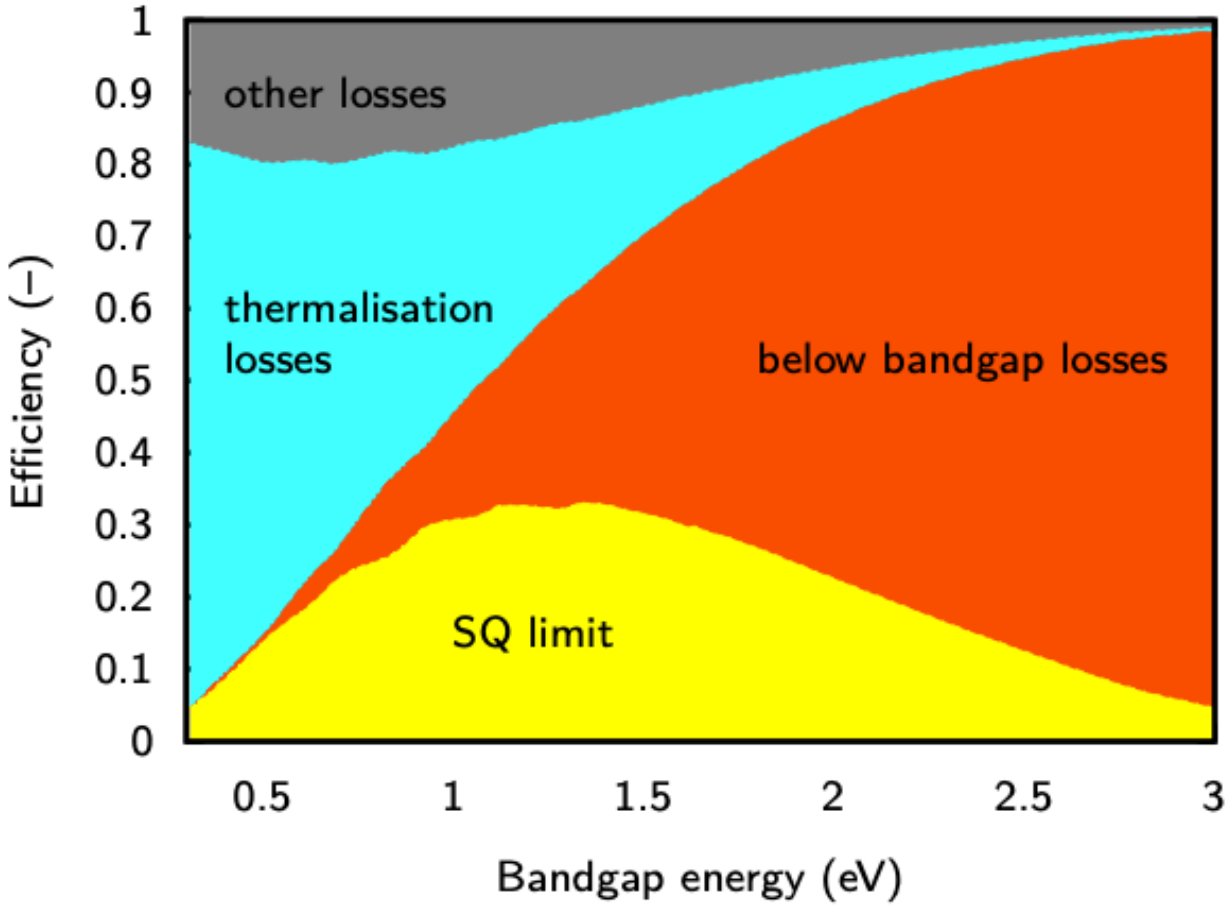




Design Aspects: Losses and Efficiency

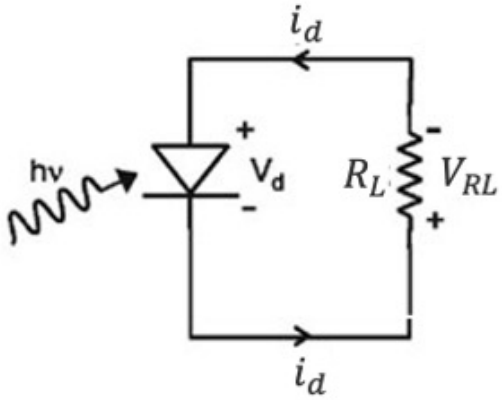
Factors affecting Open-circuit voltage V_{oc} and Short-circuit current I_{sc}.

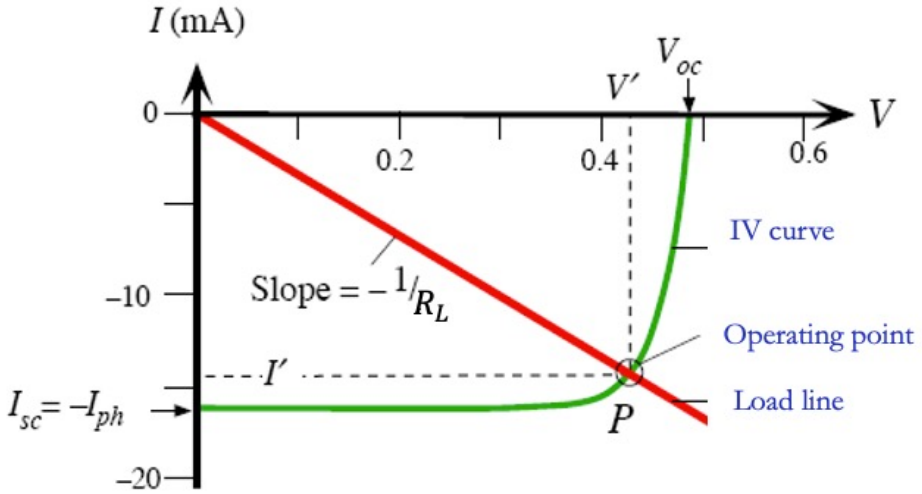
Power Budget



Solar Cell Circuit Analysis

To find the operating point, we draw the circuit load line.





Open-circuit voltage, V_{oc} , is the maximum voltage the cell can produce with no load. It is given by

$$V_{oc} = \frac{kT}{q}\ln(1 + i_p/i_s).$$

Short-circuit current, I_{sc}, is the maximum current the cell can produce when short-circuited. The small series resistance can reduce it.

The fill factor is defined as

$$FF = \frac{P_{cell}}{P_{max}} = \frac{I'\,V'}{I_{sc}\,V_{oc}}$$

The conversion efficiency

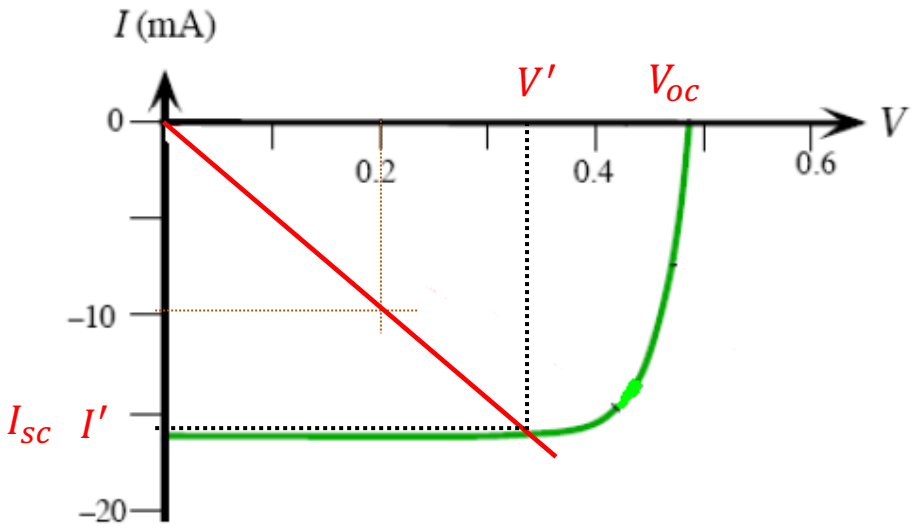
$$\eta = \frac{P_{cell}}{P_{in}} = \frac{I'\,V'}{P_{in}} = \frac{FF\,I_{sc}\,V_{oc}}{P_{in}}$$

Example

The IV curve of a certain Si solar cell with a total cell area of 4 cm^2 is given in the figure.

The cell is under illumination of 600 Wm^{-2}, and is connected to drive a load of R=20 Ω.

Calculate the fill factor and the efficiency of this solar cell?



$$FF = \frac{(16\,mA)\,(0.33)}{(16\,mA)(0.48)} = 69\,\%$$

$$\eta = \frac{(16\,mA)\,(0.33)}{(600\,W/m^2)(4\times10^{-4}m^2)} = 2.2\,\%$$

What should the load be to obtain maximum power transfer from the solar cell to the load?

Q & A

Send me an email if you have any questions.